Jishnu S Nair

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Summary

Senior Machine Learning Engineer with **6+** years of experience specializing in LLM training, reinforcement learning, and agentic AI systems. Experienced in building and deploying production-grade ML systems at enterprise scale. Published researcher with work accepted at **tier-1 AI conferences** and patents to credit.

Publications

[1] **Do Enterprise Systems Need Learned World Models? The Importance of Context to Infer Dynamics**
arXiv preprint, May 2026 — `arXiv:2605.12178`

[2] **EnterpriseOps-Gym: Environments and Evaluations for Stateful Agentic Planning and Tool Use in Enterprise Settings**
ICML 2026, March 2026 — `arXiv:2603.13594`

[3] **Apriel-Nemotron-15B-Thinker**
arXiv preprint, August 2025 — `arXiv:2508.10948`

[4] **DNR Bench: Benchmarking Over-Reasoning in Reasoning LLMs**
AAAI 2026 (Oral), March 2025 — `arXiv:2503.15793`

Experience

ServiceNow — **March 2024 – Present**
Senior Machine Learning Engineer — *Hyderabad, India*

- **Enterprise World Models**: Built learned dynamics models for enterprise agents to enable safe, predictive decision-making before action execution.
- **Reinforcement Learning for Agents**: Optimized large language models for agentic applications using advanced RL algorithms (GRPO, DAPO, PPO), achieving improved decision-making and task efficiency.
- **Agentic Orchestration**: Designed and implemented a multi-agent orchestration framework with **LangGraph** to coordinate complex tasks and automate enterprise-scale workflows.
- **Synthetic Data Generation Framework**: Built a graph-based framework leveraging **LangGraph** to generate high-quality synthetic datasets for Supervised Fine-Tuning (SFT) and Reinforcement Learning pipelines.
- **Reasoning Benchmark**: Developed a standardised benchmark to evaluate reasoning capabilities of LLMs across diverse tasks and scenarios.
- **Evaluation Framework**: Designed and developed a holistic evaluation framework using LangChain to analyse various capabilities and traits of LLMs by incorporating industry-standard datasets and metrics.

ServiceNow — **September 2022 – February 2024**
Machine Learning Engineer — *Hyderabad, India*

- **LLM Fine-tuning**: Fine-tuned and evaluated large language models like **StarCoder, StarCoderPlus, Llama 2** to perform tasks like text summarization, multi-turn Q&A and code generation specific to ServiceNow.
- **Inference Pipeline**: Developed an inference pipeline for large language models (LLMs) using **vLLM, TGI, and TensorRT**, incorporating model quantization and continuous batching
- **Similarity Service**: Developed and architected an enterprise-level **Similarity Search** service using the state-of-the-art Transformer-based models, ONNX, **HNSW** and integrated NLP algorithms for accurate similarity analysis.
- **Clustering Service**: Built an efficient **KMeans Clustering** Service with prefiltering capabilities using **FAISS**, enabling hassle-free clustering of large datasets at an enterprise level.
- **Mentorship**: Provided mentorship and guidance to **two interns**, resulting in the successful completion of their internship projects.

ServiceNow — **March 2022 – August 2022**
Software Engineer - AI/ML — *Hyderabad, India*

- **ML Automation**: Developed and deployed ML-based automation to identify customer issues that could be resolved by ChatBot, deflecting **1.75M USD** worth of tickets from human resolution every quarter.

ServiceNow **July 2020 – February 2022**
Associate Software Engineer - AI/ML *Hyderabad, India*

- **Semantic Search**: Built and deployed semantic search service using the RoBERTa algorithm to retrieve relevant articles for customer incident tickets, resolving **1.35M USD** worth of tickets per year without human intervention.
- **Virtual Agent**: Created and maintained virtual agent topics using NLU (Natural Language Understanding), thereby improving customer experience and reducing costs.

DBS Partners **March 2019 – October 2019**
Machine Learning Intern *Remote, India*

- **Contract Analytics**: Developed and deployed NLP models to extract relevant features from employment contracts using custom datasets and advanced NLP algorithms, such as **RNN, LSTM**, and **RASA NLU.**

Technical Skills

Languages: Python, SQL, JavaScript, Java
ML & RL: PPO, GRPO, DAPO, SFT, DPO, RLHF, BERT, RoBERTa, Transformers, Diffusion Models, RNN, LSTM
LLM & Agentic: LangChain, LangGraph, vLLM, TensorRT, DeepSpeed, ONNX. TensorFlow, PyTorch
Evaluation & Benchmarking: LLM-as-Judge, Reward Modeling, Benchmark Design, Synthetic Data Generation
Infrastructure: FAISS, Docker, Git, Azure, H100/GPU Clusters, NCCL, Distributed Training, Django, Flask

Education

Government Model Engineering College, Kochi **2016 – 2020**
Bachelor of Technology in Computer Science and Engineering *Kochi, India*
CGPA : 8.8/10

Patents

Smart Persistence Of Model For Effective Predictions And Updates **January 2026**
US Patent No. 18/626,139 *Patent Issued*

Platform Agnostic Scalable And High-performance Semantic Search Framework **September 2025**
US Patent No. 18/623,844 *Patent Issued*

Framework Neutral And Updatable Clustering Model **April 2024**
US Patent Application No. 18/426,810 *Patent Filed*

Awards and Recognition

ATG Llama Award: Recognized as a top contributor to the field of machine learning by ServiceNow, receiving the prestigious Llama Award **twice (2022 and 2023)** with a recognition ratio of just **14%**.